

NeuroKey Multi-Channel Electrophysiological Signal Acquisition System Software User Manual V1.0

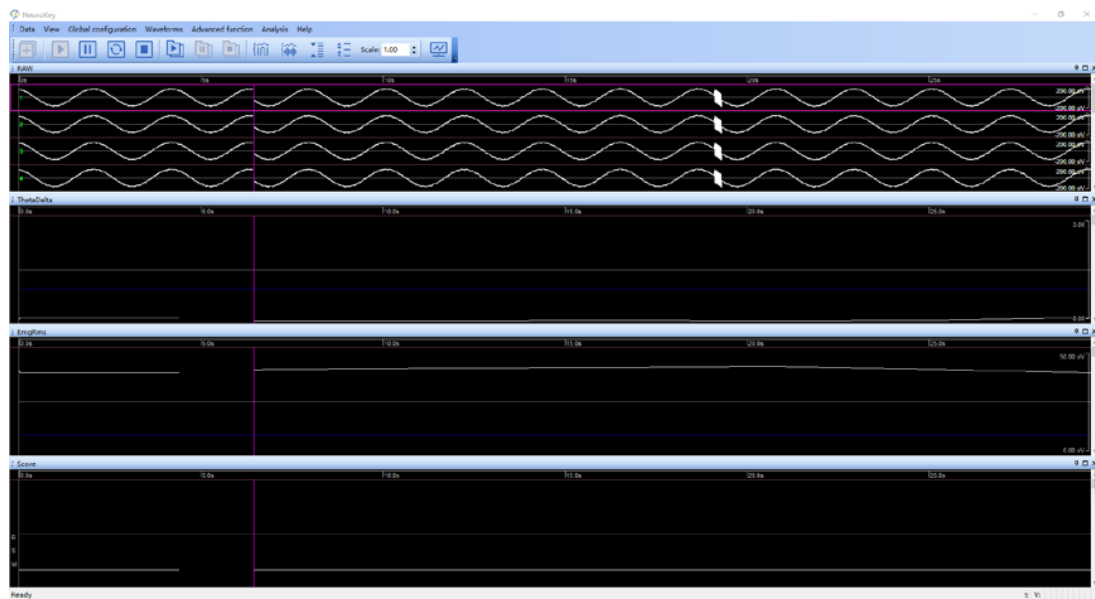


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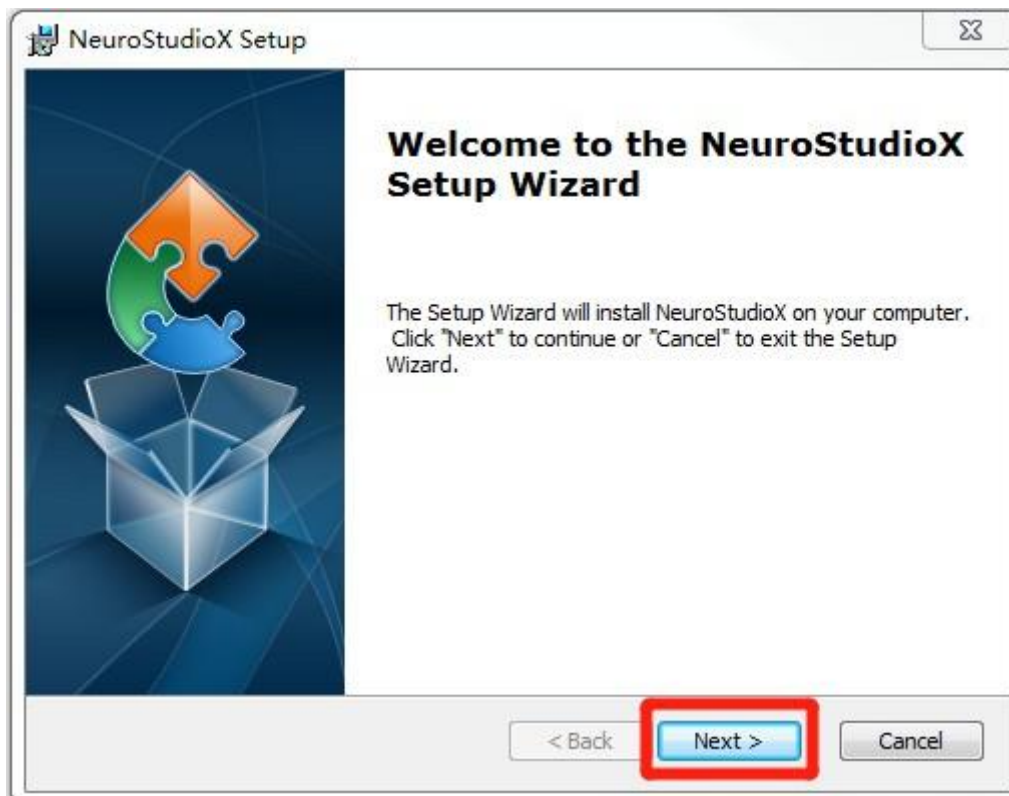
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1 Software Installation

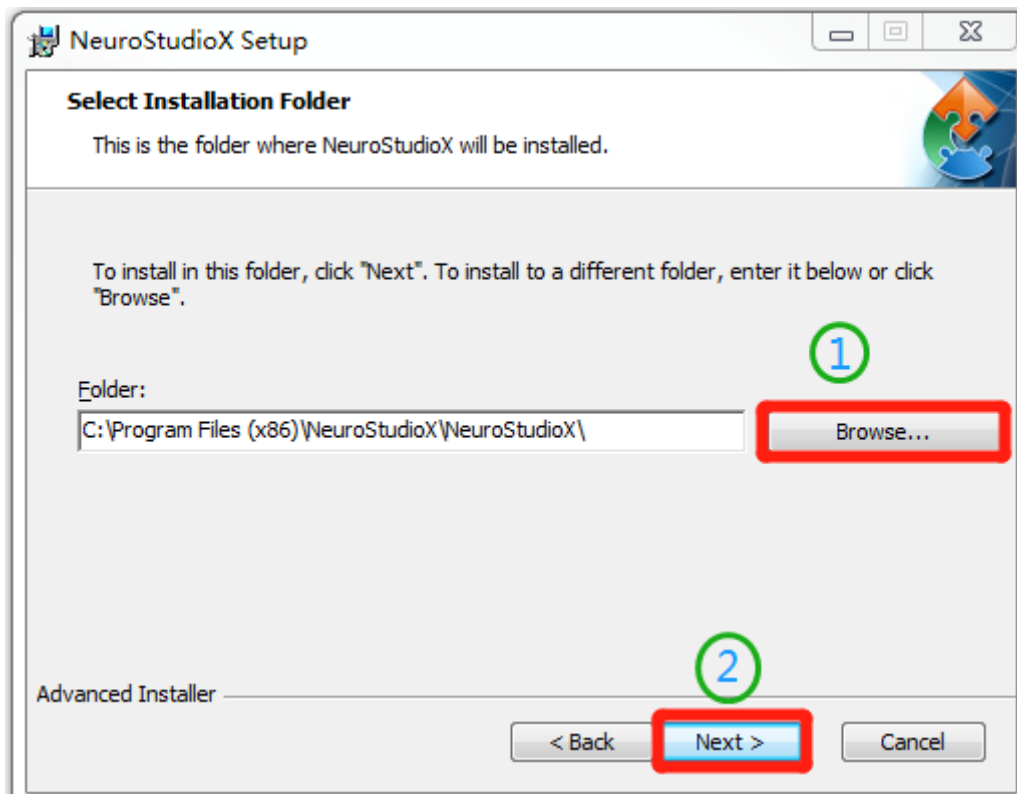
Locate the installation package and find **NeuroKeyV1.0.msi**.

 [NeuroKeyV1.0.msi](#)

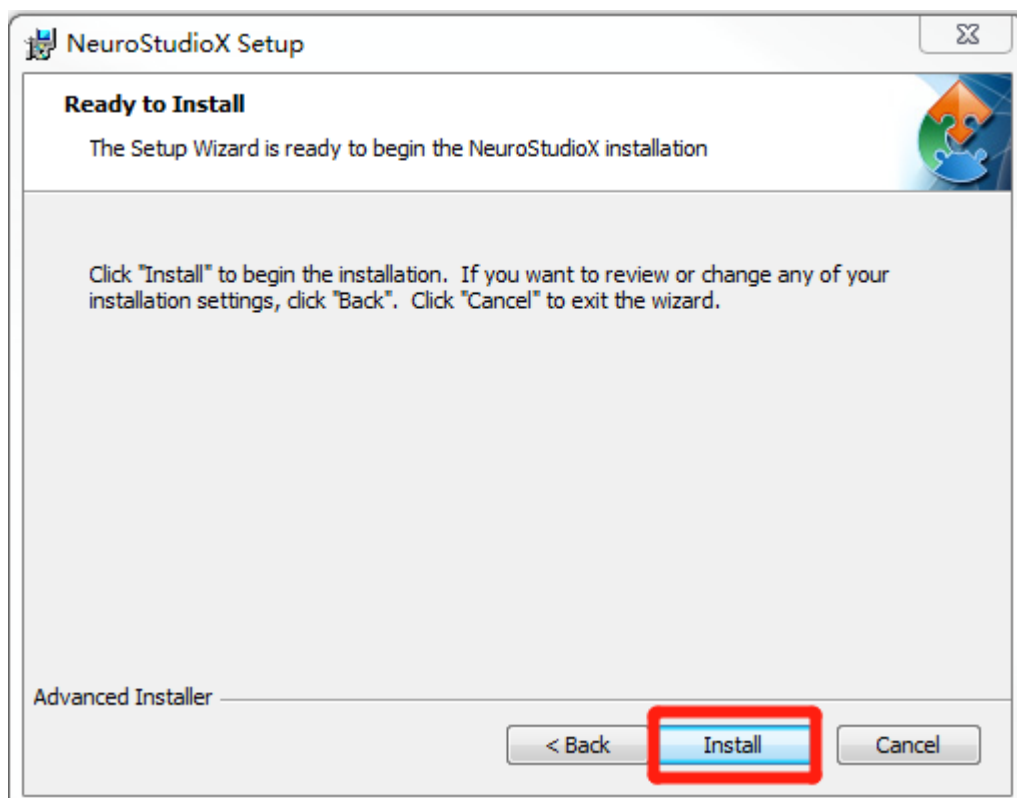
Double-click **NeuroKeyV1.0.msi** to start installation.



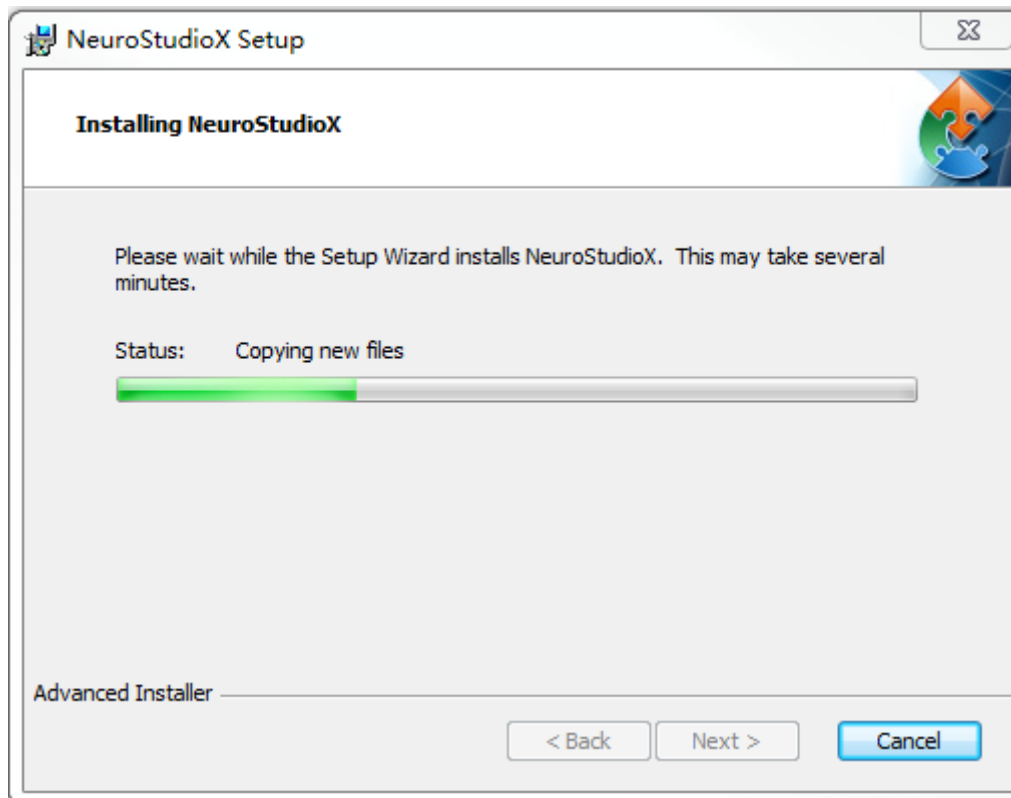
Click **Next** to proceed:



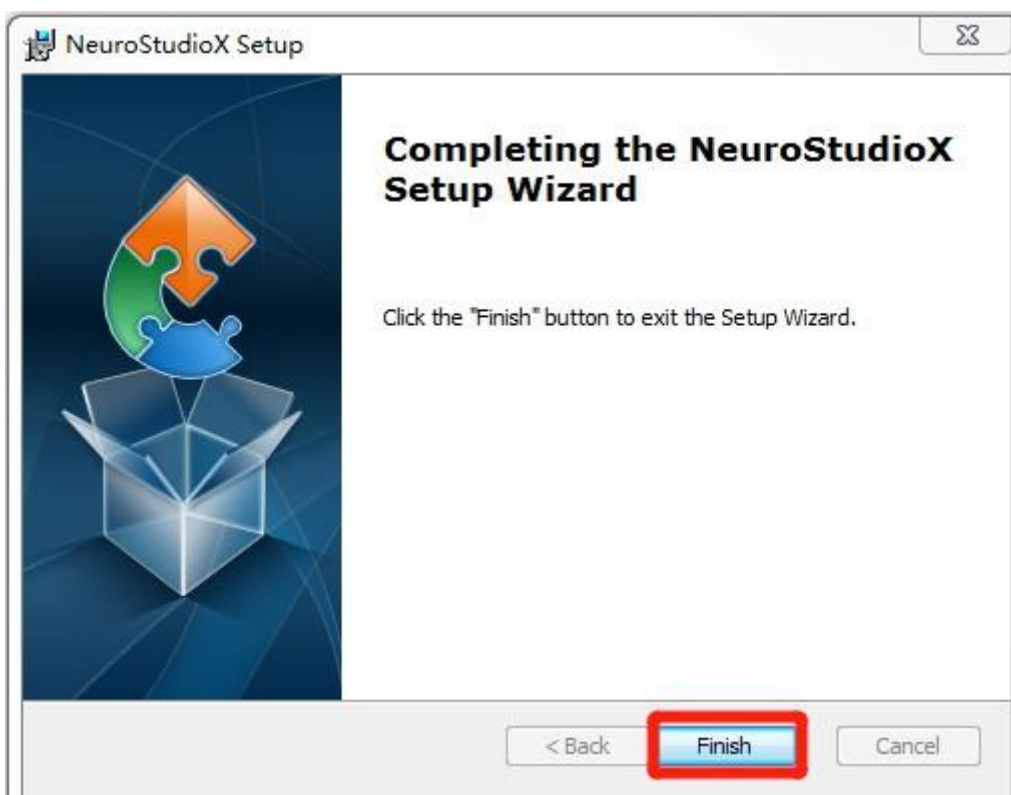
- ① You may change the installation directory
- ② Click **Next** to continue:



Click **Install** to begin installation:



Wait while the installation is in progress:

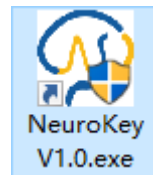


Click **Finish** to complete the installation.

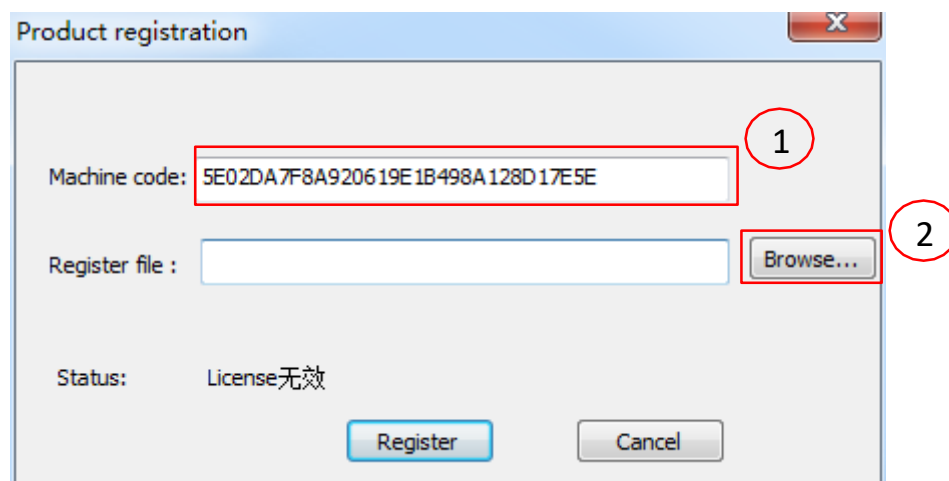
2 Basic Operation

2.1 Software Registration

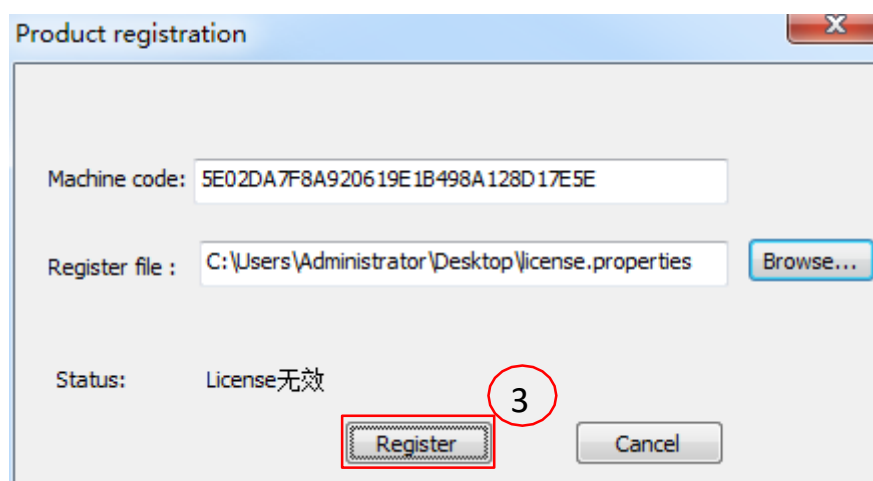
After installation, a shortcut for **NeuroKeyV1.0** will appear on the desktop:



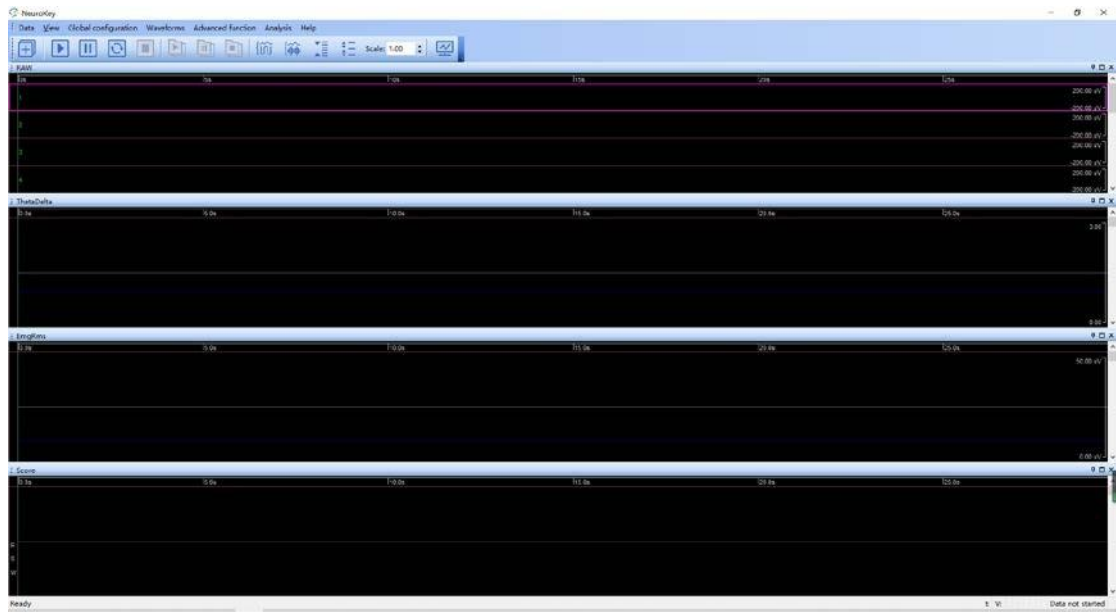
When launching the software for the first time, license registration is required. The following dialog will appear:



1. The system generates a unique **Request Code** for each computer. Copy this code and email it to **xzz@greathink.com**, including your **name** and **company/institution**. After approval, you will receive a **license file**.
2. Save the license file locally, click **Browse** to select it, and click **Register** to complete registration.



The main software interface will then appear:



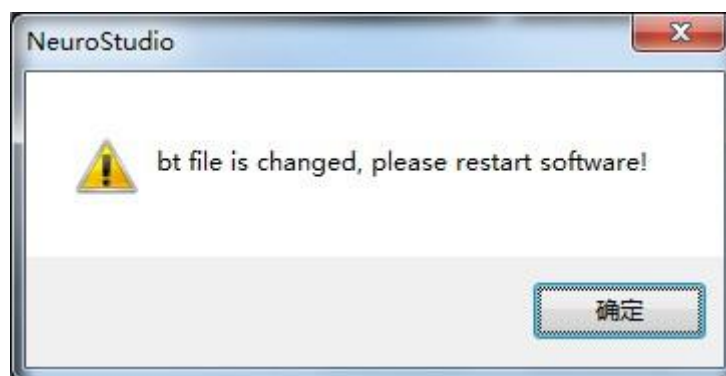
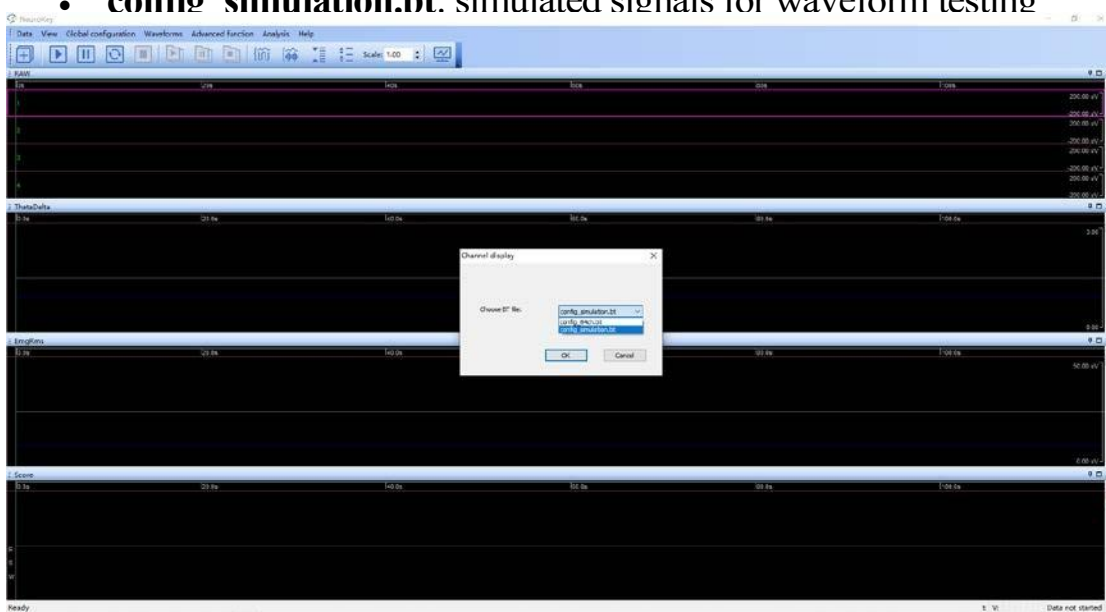
If your interface layout differs from the default, select **View** → **Default** to restore the default layout. This action does **not** affect data acquisition.

2.2 Software Configuration

Simulation signals are used here for demonstration.

Select **Global Configuration** → **Channel Display** → **config_simulation.bt** to load simulated signals.

- **config_64.bt**: real signal acquisition display (up to 64 channels)
- **config_simulation.bt**: simulated signals for waveform testing



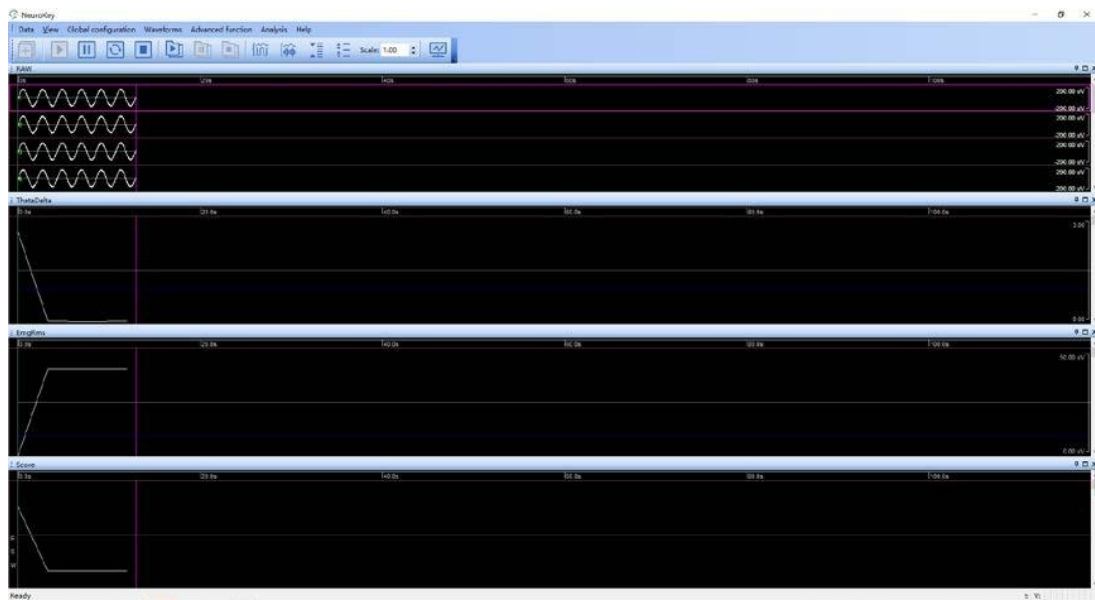
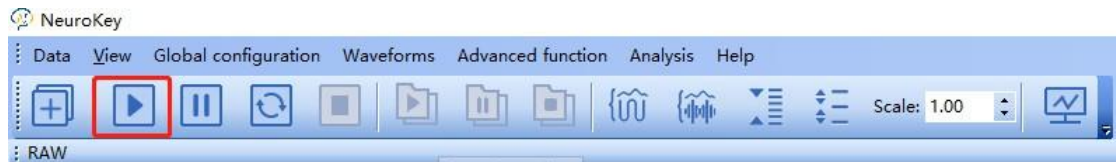
Click **OK**.

A prompt will appear: *“The bt file has changed. Please restart the software.”*
Click **OK** and restart the software .

2.3 Signal Acquisition

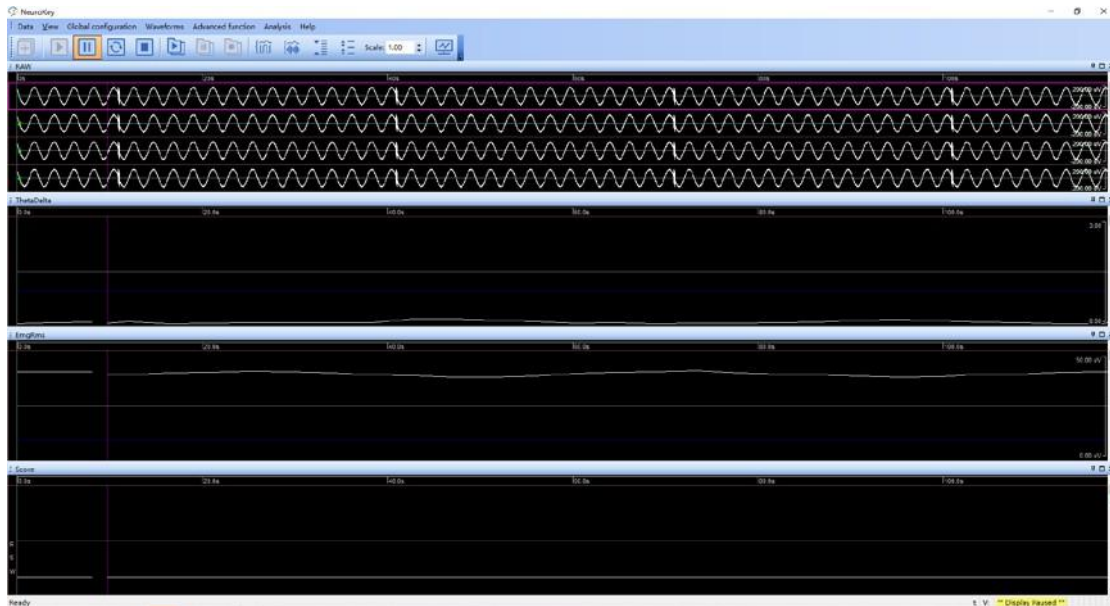
2.3.1 Start Data

Click **Data** → **Start Data** or the **Start** button on the toolbar to begin acquisition:



2.3.2 Display Pause

Click **Data** → **Display Pause** or the **Pause** button:



The display pauses, and “**Display Paused**” appears in yellow at the bottom right.

Note: Display pause does **not** stop data acquisition or recording. Recording time and file size continue updating.

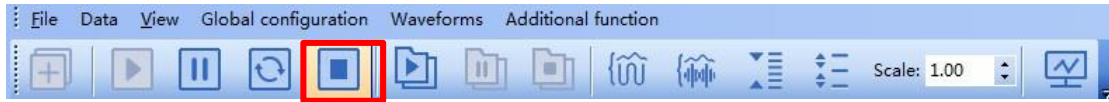
2.3.3 Refresh

Click **Data** → **Refresh** or the **Refresh** toolbar button to manually refresh all waveform windows:



2.3.4 Stop Data

Click **Data** → **Stop Data** or the **Stop** toolbar button:



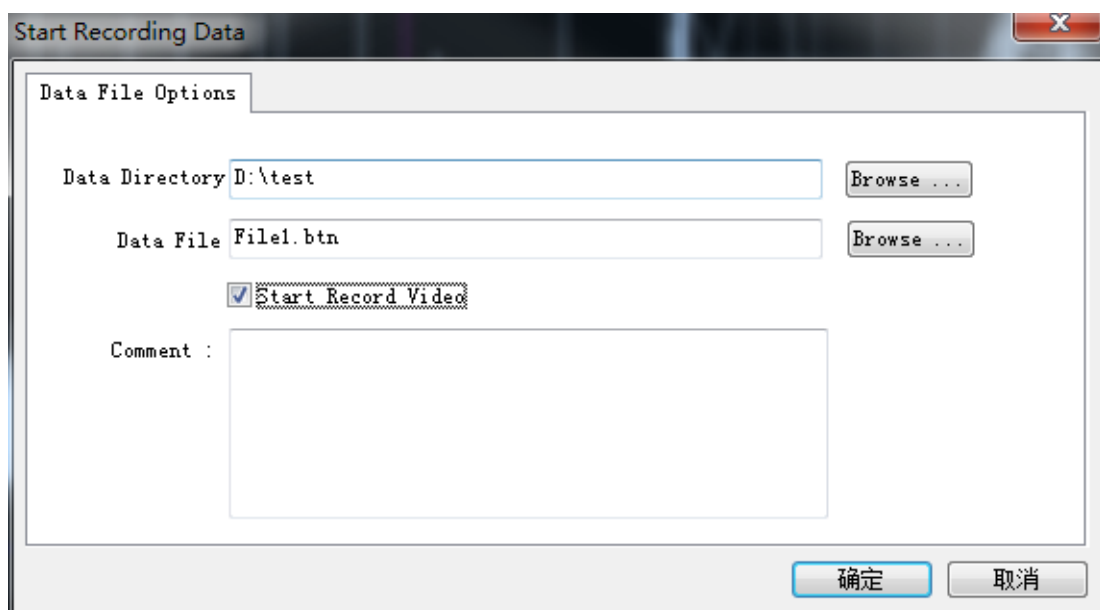
Acquisition stops, but all parameter settings (e.g., thresholds) are preserved for the next run.

2.4 Data Recording

2.4.1 Start Recording

Before recording, configure saved channels (see Section [3.4.1](#)

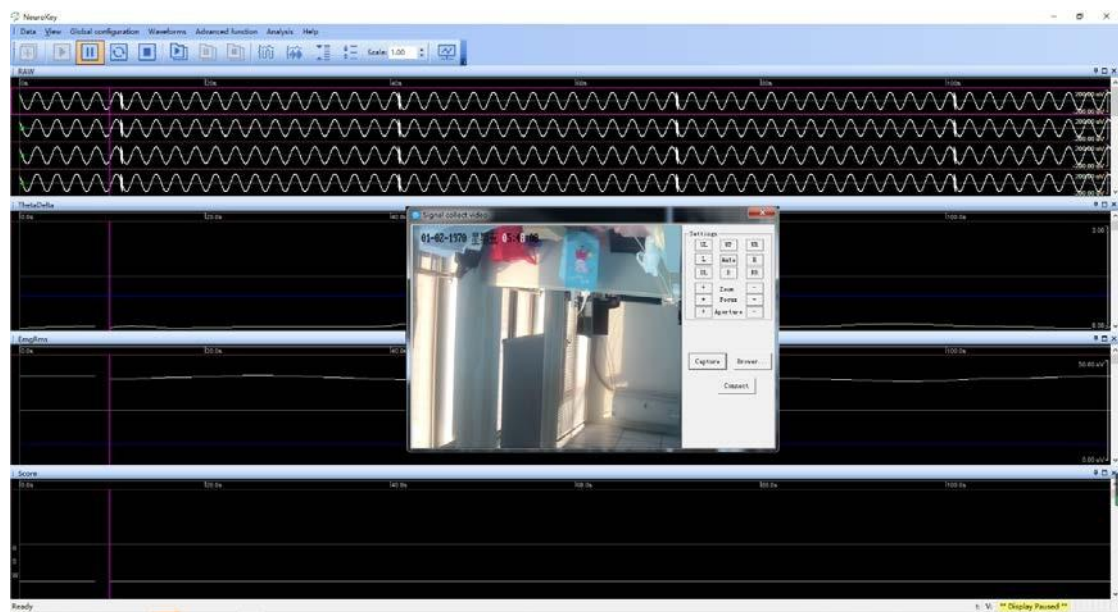
[Channel Save Select](#)). Start Recording:



Configure:

- **Data Directory:** recommended to create a new folder on drive D
- **Data File:** file name
- **Comment:** user notes
- **Start Record Video:** enables synchronized video recording

Click **OK** to start recording.



Recording duration and file size are displayed in real time at the bottom right.

2.4.2 Pause Recording

Click **Pause Recording** to temporarily stop saving.:



The recording time display turns yellow. Click again to resume.

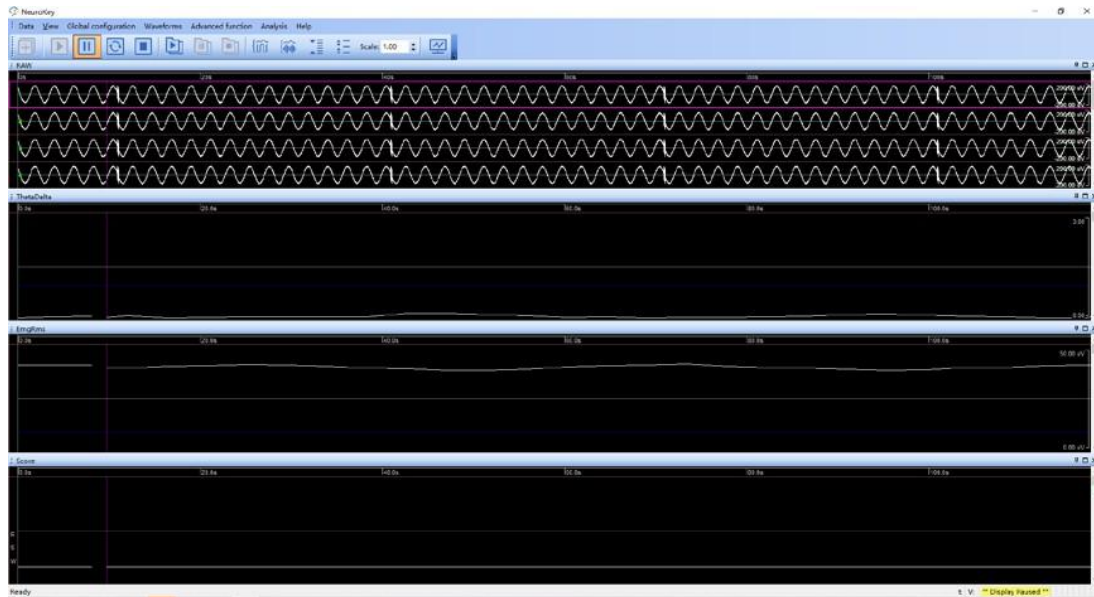
2.4.3 Stop Recording

Click **Stop Recording** to end data saving:

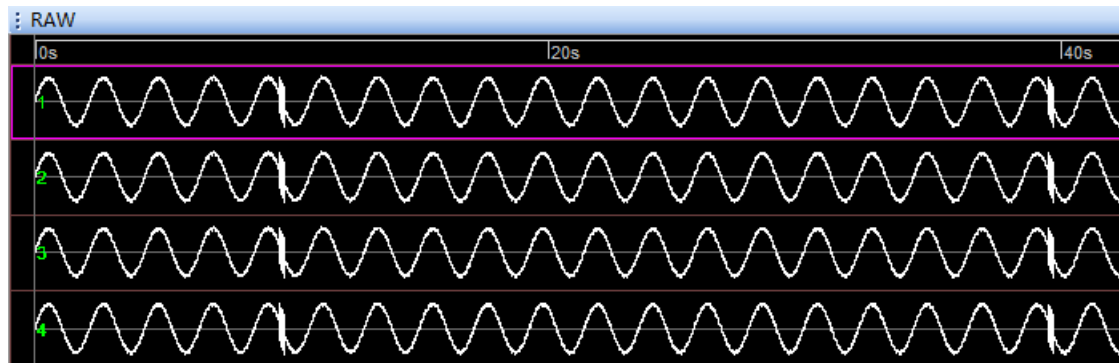


2.5 Waveform Windows

The **RAW** window displays original signals acquired by the hardware.



2.5.1 RAW



The **RAW** window displays original signals acquired by the hardware.

- A vertical colored scan line shows the current position
- The scan line loops from right to left, overwriting old data
- The top axis represents relative time

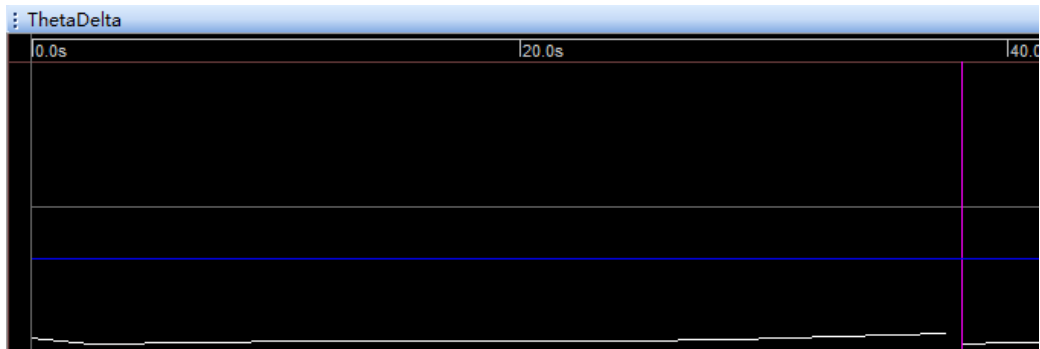
Toolbar functions:

- **Refresh**: refresh waveform
- **Speed Up / Speed Down**: adjust scan speed
- **Scale Up / Scale Down**: adjust number of displayed channels
- **Scale value**: adjust amplitude (default = 1)



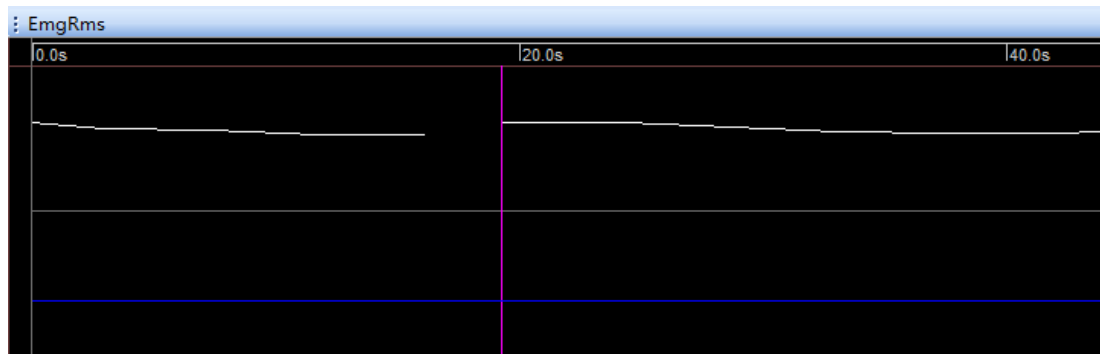
2.5.2 Theta/Delta

EEG signals contain alpha (α), beta (β), theta (θ), and delta (δ) waves. The **Theta/Delta ratio** is used to classify sleep states in small animals. Drag the blue threshold line to adjust parameters in real time.



2.5.3 EmgRms

EEG signals contain alpha (α), beta (β), theta (θ), and delta (δ) waves. The **Theta/Delta ratio** is used to classify sleep states in small animals. Drag the blue threshold line to adjust parameters in real time.

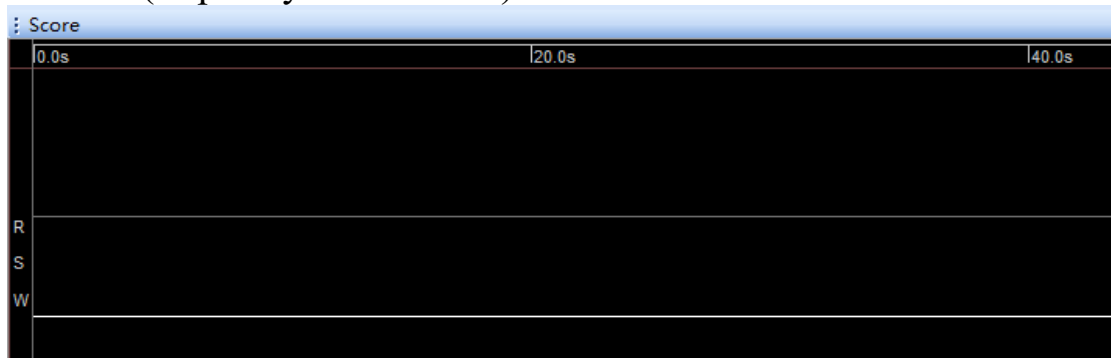


2.5.4 Score

Configured via **Analysis** → **SleepScore**.

Displays real-time classification:

- Sleep
- Awake
- REM (Rapid Eye Movement)



3 Menu Bar Functions

3.1 File

3.1.1 Open

Used for playback of **.btn** files (Ctrl+O).

Playback is only available when acquisition is stopped.



Channel Display settings must match those used during recording.

3.2 Data

3.2.1 General Operation

软件菜单栏 Data 下的选项分别对应工具栏中的按钮，即如下：



Start Data: 

Display Pause: 

Refresh: 

stop data: 

Start Recording: 

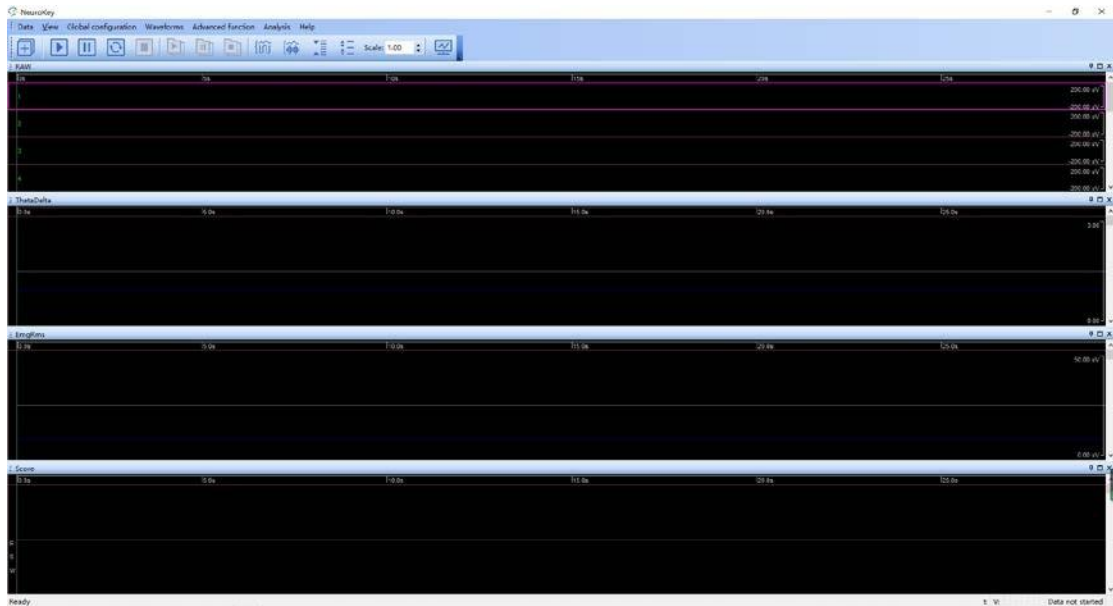
Pause Recording: 

Stop Recording: 

3.3 View

3.3.1 Default

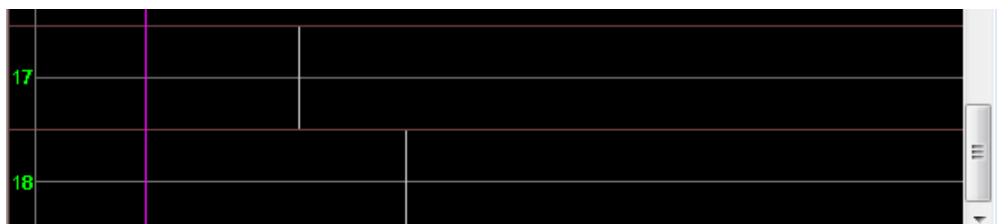
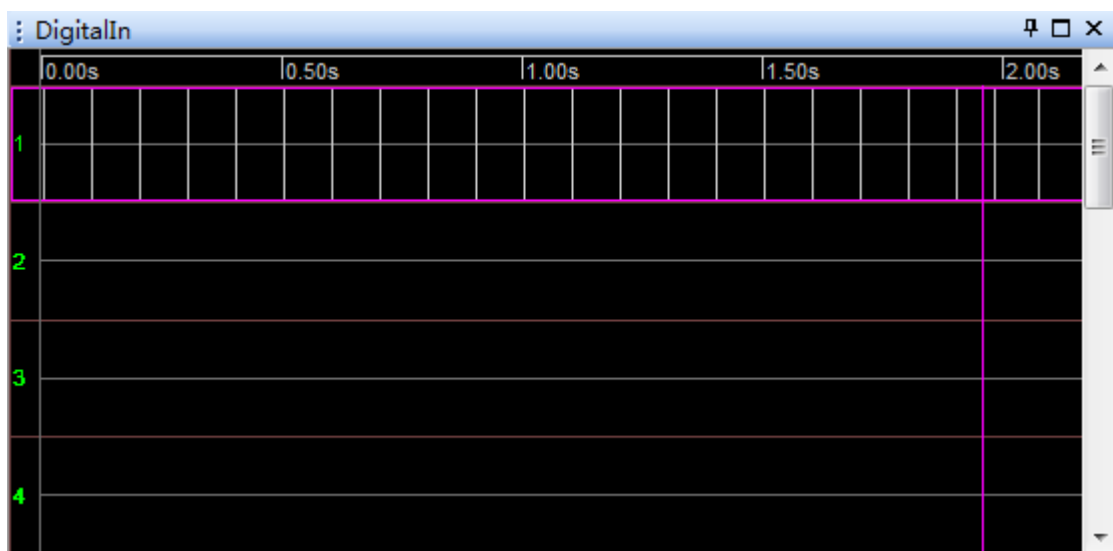
Restores the default interface layout after resizing or rearranging



3.3.2 Digital IN

Used for event marking via auxiliary digital inputs.

- High-level input (3.3–5 V, >3 ms) generates a vertical marker
- Markers are saved automatically during recording
- Keyboard marking:
 - **Ctrl+1** → Digital IN channel 17
 - **Ctrl+2** → Digital IN channel 18

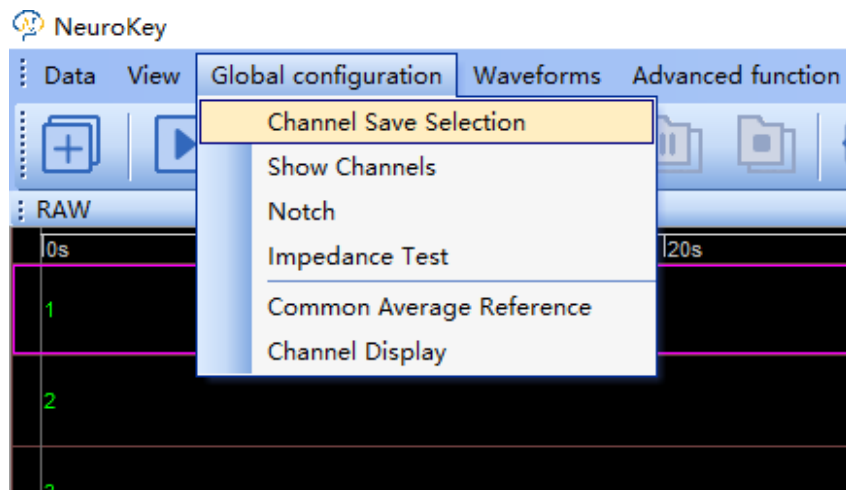


3.4 Global Configuration

3.4.1 Channel Save Selection

Configure which channels are saved:

- **WB-Continuous:** select signal channels (Ctrl / Shift supported)
- **Analog:** auxiliary analog inputs
- **DigitalIn:** digital marker channels
- **Configuration:** segmented recording duration



Save Configuration

WB-Continuous Analog DigitalIn Configuration

Name	Continuous
CH.01	☒
CH.02	☒
CH.03	☒
CH.04	☒
CH.05	☒
CH.06	☒
CH.07	☒
CH.08	☒
CH.09	☒
CH.10	☒
CH.11	☒
CH.12	☒
CH.13	☒
CH.14	☒
CH.15	☒
CH.16	☒
CH.17	☒

SelectAll SelectContinus

OK Cancel

Save Configuration

WB-Continuous Analog DigitalIn Configuration

Name	Analog	Remark
Analog01	☐	
Analog02	☐	
Analog03	☐	
Analog04	☐	
Analog05	☐	
Analog06	☐	
Analog07	☐	
Analog08	☐	

OK Cancel

Save Configuration

WB-Continuous Analog DigitalIn Configuration

Name	DigitalIn	Remark
DigitalIn01	☒	
DigitalIn02	☒	
DigitalIn03	☐	
DigitalIn04	☐	
DigitalIn05	☐	
DigitalIn06	☐	
DigitalIn07	☐	
DigitalIn08	☐	
DigitalIn09	☐	
DigitalIn10	☐	
DigitalIn11	☐	
DigitalIn12	☐	
DigitalIn13	☐	
DigitalIn14	☐	
DigitalIn15	☐	
DigitalIn16	☐	

OK Cancel

Save Configuration

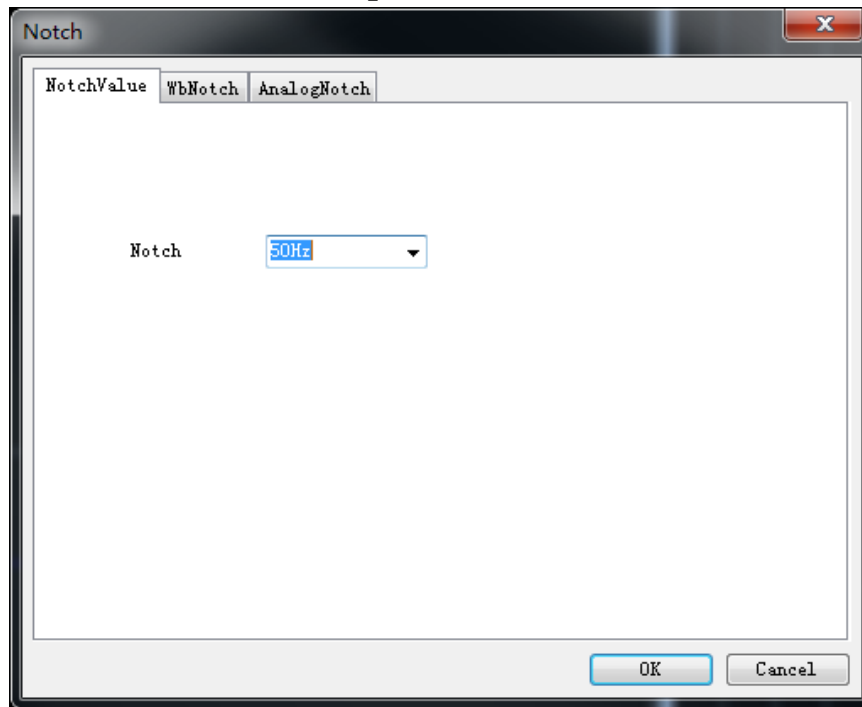
WB-Continuous Analog DigitalIn Configuration

Segment save time: 9999999 min

OK Cancel

3.4.2 Notch

Removes **50 Hz or 60 Hz** power-line interference:



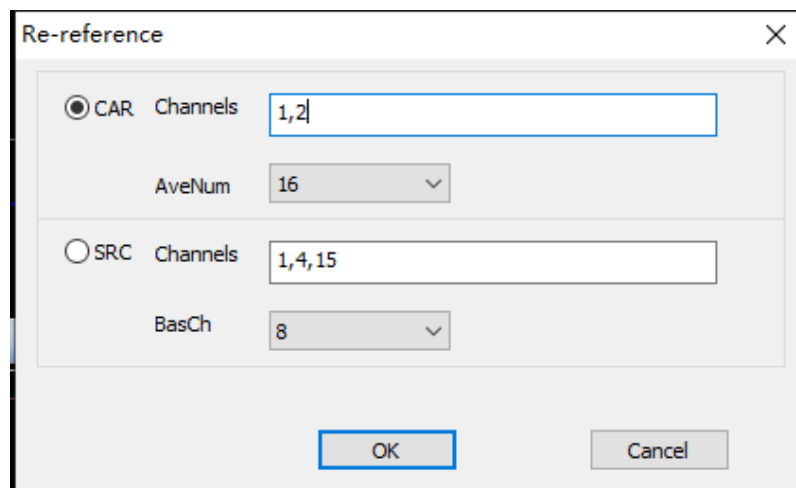
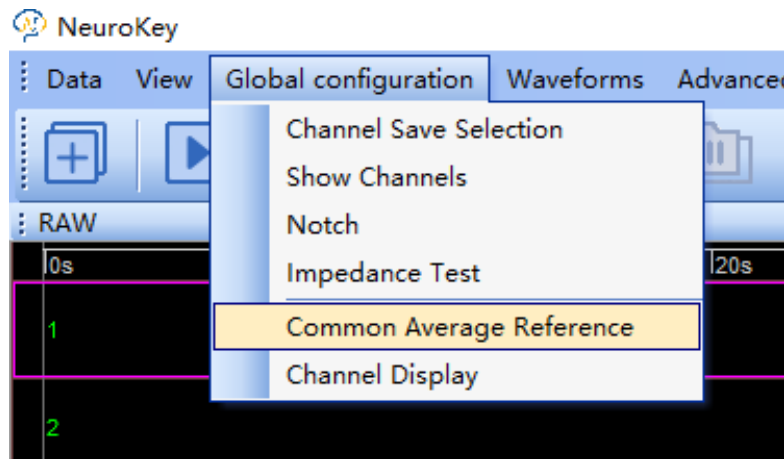
3.4.3 Impedance

Electrode impedance testing
(Supported on NeuroStudio devices only).

3.4.4 Common Average Reference(CAR)

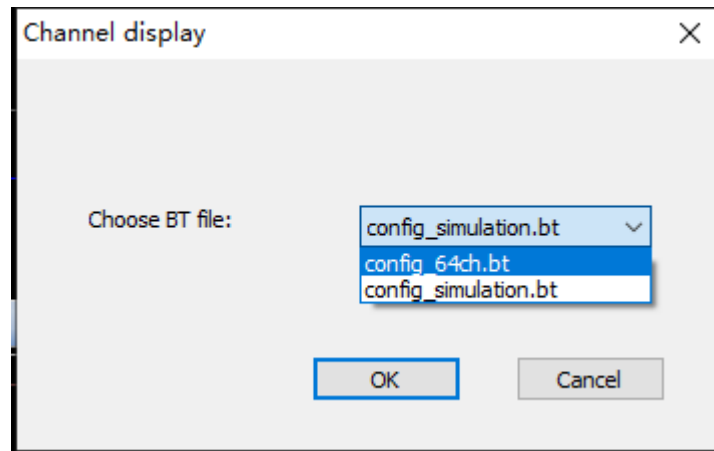
CAR subtracts the average of selected channels to reduce common noise.

- Enable CAR by checking the box
 - Channels must be separated by commas (English input)
- SCR mode** subtracts a reference channel (**BasicCh**) from selected channels.

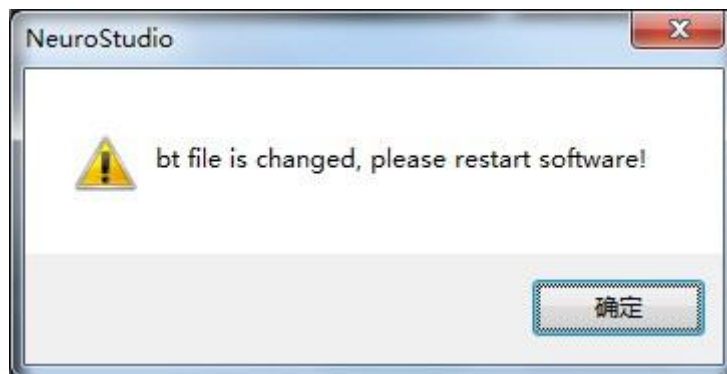


3.4.5 Channel Display

Select **config_64.bt** or **config_simulation.bt**.



Restart software after changing configuration.



Adjust scan speed, number of displayed windows, and amplitude scaling. Minimum scale value: **0.02**

3.5 Waveforms

3.5.1 Waveforms



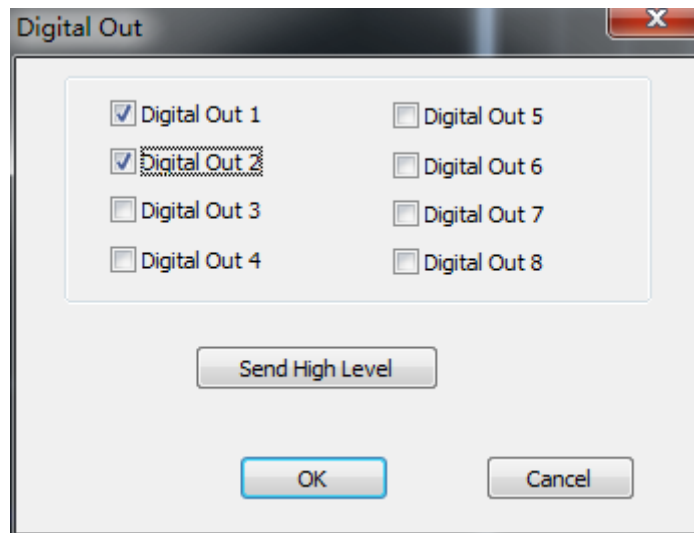
Speed up/down Scale up/down

3.6 Additional Function

3.6.1 Analog Out

(NeuroStudio only)。

3.6.2 Digital Out



Manually output a TTL pulse (10 ms) for synchronization with external devices:

3.7 Analysis

3.7.1 SleepScore

OnlineSleepScore performs real-time sleep state analysis。



The screenshot shows a software window titled "Online SleepScore" with a close button (X) in the top right corner. The window contains several configuration options:

- BANK Select:** A dropdown menu currently set to "BankA".
- EEG Channel:** A dropdown menu currently set to "CH1".
- EMG Channel:** A dropdown menu currently set to "CH2".
- 时间片:** A dropdown menu currently set to "4s".
- ThetaDelta阈值:** A text input field containing the value "0.96".
- EmgRms阈值:** A text input field containing the value "9.47".
- 睡眠剥夺:** A checkbox that is currently unchecked.
- 剥夺在:** A dropdown menu currently set to "Sleep".
- sleepdeprivation文件:** A text input field with a "选择" (Select) button next to it.
- Buttons:** At the bottom, there are two buttons: "Save&Exit" (highlighted with a blue dashed border) and "Cancel".

Parameters:

- Theta/Delta threshold
- EmgRms threshold
- Time window: 4 s / 10 s / 30 s

BANK mapping:

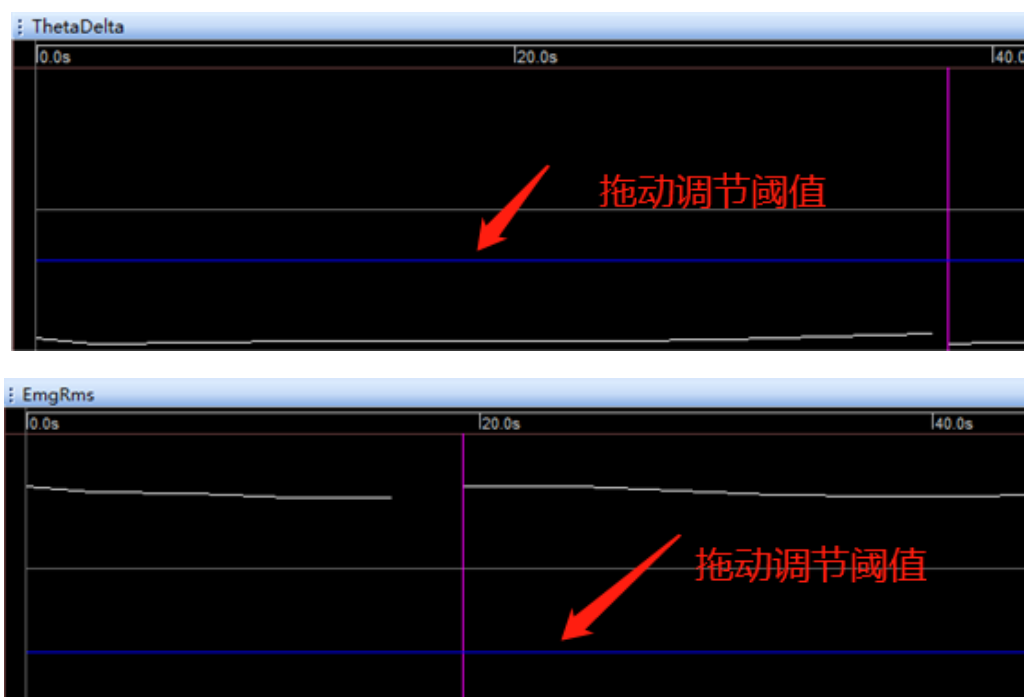
- BANKA: Ch 1–4

- BANKB: Ch 5–8
- BANKC: Ch 9–12
- BANKD: Ch 13–16

Outputs:

- Sleep
- Awake
- REM

Supports synchronization with sleep deprivation devices.



Online SleepScore

BANK Select: BankA

EEG Channel: CH1

EMG Channel: CH2

时间片: 4s

ThetaDelta阈值: 0.96

EmgRms阈值: 9.47

☐ 睡眠剥夺

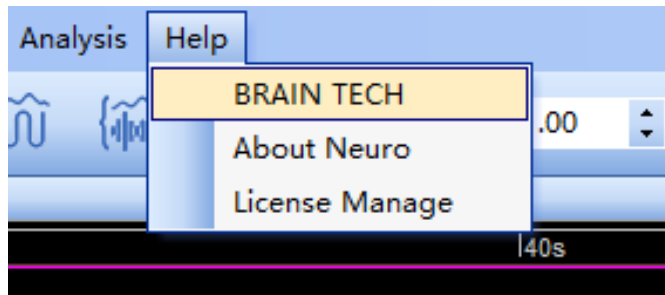
剥夺在: Sleep

sleepdeprivation文件: 选择

Save&Exit Cancel

3.8 Help

3.8.1 BRAIN TECH

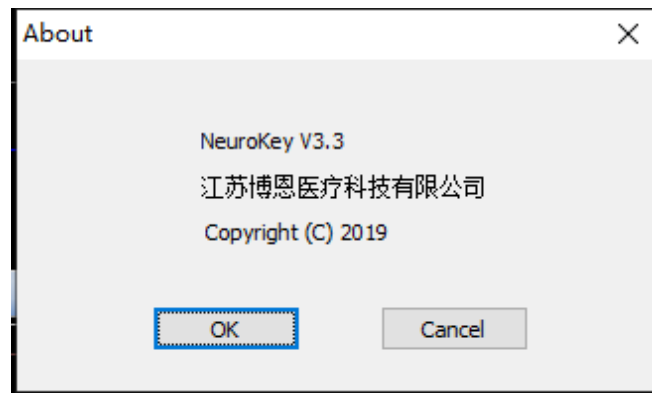


Opens official website:



<http://www.brainmedtech.com/>

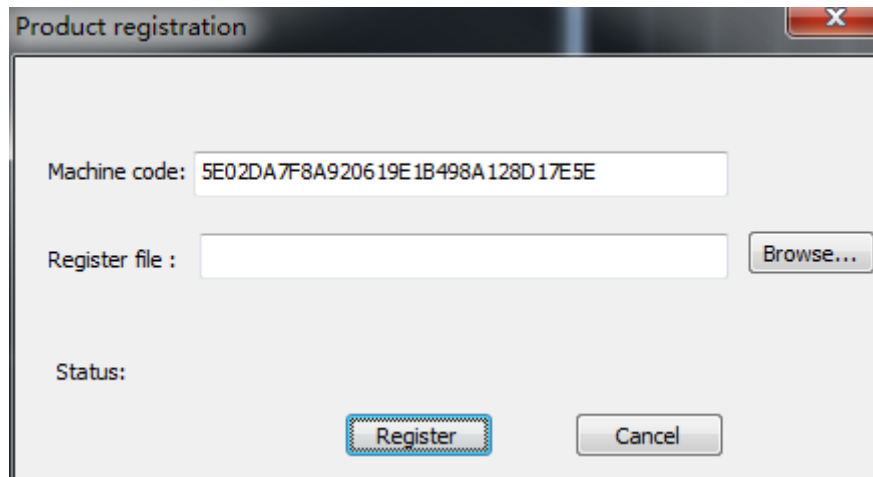
3.8.2 About Neuro



Displays software version information.

3.8.3 License Manage

Opens license registration window.



The image shows a 'Product registration' dialog box with a title bar containing a close button (X). The dialog contains the following elements:

- A label 'Machine code:' followed by a text input field containing the value '5E02DA7F8A920619E1B498A128D17E5E'.
- A label 'Register file :' followed by an empty text input field and a 'Browse...' button to its right.
- A label 'Status:' located below the input fields.
- Two buttons at the bottom: 'Register' (highlighted with a blue dashed border) and 'Cancel'.